

PolyLoop: Technical Architecture, Engineering Blueprint, and Operational Strategy for Automated E-Waste Polymer Identification and Upcycling Theoretical Foundations and Chemistry of E-Waste Plastics

Electronic waste represents the fastest-growing domestic waste stream in the modern world¹. While established metallurgical refining infrastructure recovers precious metals such as gold, silver, copper, and palladium from printed circuit boards, the polymeric housings that enclose these assemblies are overwhelmingly discarded¹. Polymeric casings account for approximately 20% to 30% of total e-waste by mass, yet over 80% of this material is routed to landfills or incinerators¹. This systemic leakage in circular materials management stems from two intractable technical barriers: macromolecular thermodynamic incompatibility and persistent chemical contamination from legacy halogenated additives¹.

Thermodynamic Polymer Incompatibility

Post-consumer electronics rely predominantly on styrenic polymers and engineering thermoplastics, with market volumes dominated by Acrylonitrile Butadiene Styrene (ABS), High-Impact Polystyrene (HIPS), Polycarbonate (PC), and PC-ABS blends¹. Because these polymers are molded into black, dark gray, or beige casings, visual identification is nearly impossible¹. However, commingling these resins during mechanical melt reprocessing causes acute mechanical failure¹.

The physical mechanism of this breakdown is described by the Flory-Huggins theory of polymer solutions and blends. The Gibbs free energy of mixing per unit volume for two chemically distinct macromolecules is defined by the relationship:

$$\Delta G_m = \Delta H_m - T\Delta S_m$$

Because industrial polymers possess macromolecular chain lengths with high degrees of polymerization (N), the combinatorial entropy of mixing (ΔS_m) approaches zero³. Consequently, the thermodynamic spontaneity of mixing is dictated almost entirely by the enthalpy of mixing (ΔH_m), which is directly proportional to the Flory-Huggins interaction parameter (χ_{12}):

$$\chi_{12} = \frac{V_{\text{ref}}}{k_B T} (\delta_1 - \delta_2)^2$$

In this expression, δ_1 and δ_2 represent the respective Hansen solubility parameters of the individual polymer phases, V_{ref} is the reference volume, and $k_B T$ is the thermal energy³. Because almost all chemically dissimilar polymer pairs yield positive interaction parameters ($\chi_{12} > 0$), spontaneous thermodynamic miscibility cannot occur³.

When incompatible matrices such as ABS and HIPS are melted together, the polymers undergo micro-phase separation¹. ABS consists of a polar continuous styrene-acrylonitrile (SAN) copolymer matrix grafted onto polybutadiene rubber particles³. In contrast, HIPS consists of a non-polar polystyrene continuous phase containing dispersed polybutadiene spheres³. The polar nitrile ($-C \equiv N$) dipoles in the ABS matrix repel the non-polar aromatic rings of the HIPS matrix, resulting in high interfacial tension, zero macromolecular interdiffusion, and negligible chain entanglement across the boundary³. Under external mechanical loading, these unbonded phase interfaces act as critical stress concentrators that promote micro-void nucleation, uncontrolled crazing, and catastrophic delamination¹. Contaminating an ABS matrix with as little as 3% to 5% HIPS by mass reduces notched Izod impact resistance and tensile elongation at break by more than 70%, rendering the recycled plastic too brittle for structural applications¹.

Polymer Combination	Chemical Structure Disparity	Interfacial Melt Mechanics	Resulting Mechanical Defect Mode
ABS + HIPS	Polar nitrile groups on the SAN copolymer matrix versus non-polar aromatic polystyrene ³ .	High interfacial tension; total absence of molecular chain entanglement across phase boundaries ³ .	Severe delamination under shear; loss of over 70% in impact and tensile strength ¹ .
ABS + PC	SAN grafted terpolymer mixed with aromatic bisphenol-A	Thermodynamically favorable partial miscibility; forms an engineering alloy under correct	High tensile strength and impact toughness, provided feedstock is dried to prevent

	polycarbonate ⁴ .	shear ¹ .	hydrolysis ¹ .
HIPS + PC	Non-polar styrenic hydrocarbon matrix versus polar aromatic carbonate backbone ³ .	Complete thermodynamic immiscibility; rapid macrophase segregation during cooling ³ .	Severe matrix embrittlement; filament snaps inside 3D printer drive gears ¹ .
Polyolefins (PP/PE) + Styrenics	Linear aliphatic hydrocarbon chains versus bulky aromatic/nitrile pendant groups ⁷ .	Extreme structural divergence; gross phase separation during laminar melt flow ⁷ .	Complete failure of layer-to-layer adhesion; material shears along laminar boundaries ¹ .

Hazardous Additives: Brominated Flame Retardants and Regulatory Directives

The secondary barrier preventing the reuse of e-waste plastic is chemical toxicity¹. Historically, international fire safety standards, such as UL 94 V-0, required that polymer enclosures housing power supplies, cathode-ray tubes (CRTs), and high-voltage transformers extinguish quickly when ignited⁶. Manufacturers met this requirement by adding 10% to 25% by weight of Brominated Flame Retardants (BFRs) to raw polymer formulations, typically using 3% to 8%

Antimony Trioxide (Sb_2O_3) as an inorganic synergist⁶.

The primary legacy BFR additives include Polybrominated Diphenyl Ethers (PBDEs), such as Decabromodiphenyl ether (DecaBDE) and Octabromodiphenyl ether (OctaBDE); Tetrabromobisphenol A (TBBPA), used either as an additive or reactive compound; and Hexabromocyclododecane (HBCD)⁶. Because additive flame retardants are physically dispersed rather than chemically bonded to the host polymer matrix, they can leach out over time³.

Thermally, these halogenated additives work through gas-phase radical trapping: upon exposure to flame, they cleave weak carbon-bromine ($\text{C} - \text{Br}$) bonds to generate reactive bromine radicals ($\text{Br}^\bullet$), which neutralize high-energy hydroxyl ($\text{OH}^\bullet$) and hydrogen ($\text{H}^\bullet$) free radicals in the combustion zone. However, when BFR-contaminated plastics are reprocessed in extruders or 3D printers at temperatures between 210°C and 250°C, the thermal stress causes partial dehydrohalogenation¹. This process releases toxic Polybrominated Dibenzo-p-dioxins and Dibenzofurans (PBDD/Fs), which are bioaccumulative carcinogens and endocrine disruptors¹.

Strict international environmental regulations govern these compounds:

- The European Union RoHS Directive (2011/65/EU and 2015/863/EU) establishes a strict

maximum concentration limit of 0.1% by weight (1,000 ppm) for PBDEs and polybrominated biphenyls (PBBs) in homogeneous electrical components¹.

- The Stockholm Convention on Persistent Organic Pollutants and the EU POPs Regulation (Regulation 2019/1021) limit the concentration of specific PBDE congeners in recycled plastics to 500 ppm, prohibiting their reuse in consumer goods, children's toys, or domestic 3D printing filaments¹.
- Annex VII of the WEEE Directive (2012/19/EU) requires the selective extraction and specialized disposal of all plastics containing brominated flame retardants prior to shredding or downcycling¹.

Closed-loop recycling systems require screening mechanisms to identify and isolate BFR-contaminated components, preventing hazardous halogens from re-entering secondary consumer supply chains¹.

Appliance Category	Era of Manufacture	Primary Polymer Matrix	BFR Contamination Risk Level	Regulatory Action Protocol
CRT Televisions & CRT Monitors	Pre-2006	HIPS (63%), ABS (22%) ⁴	Critical Risk (70% – 90%): High concentrations of DecaBDE and OctaBDE ⁶ .	Reject from mechanical upcycling; route to specialized high-temperature WEEE disposal ¹ .
Laser Printers & Copiers	1998 – 2012	ABS (80%), PC-ABS (13%) ⁴	High Risk (40% – 60%): Internal fuser zones frequently compound TBBPA additives ⁶ .	Require mandatory density stratification and Beilstein flame tests before approval ¹ .
Desktop PC Chassis & Bezels	2000 – 2015	ABS (50%), PC-ABS (35%), HIPS (15%) ⁴	Moderate-High Risk (30% – 45%): Dark structural bezels often contain BFRs ⁶ .	Inspect ISO 11469 mold markings for FR codes; run density validation

				tests ¹ .
Modern Flat Displays & Laptops	2016 – Present	PC-ABS (90%), Pure PC (10%) ⁴	Low Risk (<10%): Industry transition toward organophosphorus flame retardants ⁴ .	Qualified for closed-loop upcycling and additive manufacturing filament extrusion ¹ .
Keyboards, Mice & Routers	2010 – Present	ABS (>80%), HIPS (<15%) ⁴	Minimal Risk (<5%): Low internal voltage requirements eliminate need for BFRs ⁴ .	Highly suitable for direct desktop granulation, washing, drying, and filament extrusion ¹ .

Multi-Tiered Hybrid Identification and Diagnostic Architecture

Industrial recycling facilities rely on Near-Infrared (NIR) or X-ray Fluorescence (XRF) spectroscopy systems that often cost more than \$30,000, placing them out of reach for decentralized maker spaces, academic laboratories, and small workshops¹. PolyLoop addresses this economic limitation through a three-tiered hybrid identification pipeline that combines edge computer vision, automated optical character recognition, and physical-chemical diagnostics¹.

The operational workflow begins when an e-waste component is presented to a standard local webcam¹. The image passes through a component classifier that identifies the appliance type, establishing its manufacturing era and baseline hazard profile¹.

The system then zooms in to locate and read embossed resin stamps¹. If an ISO stamp is detected, the parsed polymer type and any embedded flame-retardant markings are cross-referenced with the hazard engine¹.

If the casing has been degraded or lacks legible stamps, the system automatically redirects to an interactive diagnostic wizard¹. This tool walks the operator through stepwise density-based sink-float stratification and solvent spot tests¹.

Once verified pure and free of hazardous halogens, the polymer is cleared for thermal calculation and extrusion into 3D printer filament¹. Samples flagged with high flame-retardant concentrations are rejected and routed to hazardous waste streams¹.

Stage 1: Visual Object Classification (Ultralytics YOLO)

The first stage captures visual frames using a standard webcam connected to an Ultralytics YOLOv8n (nano) inference engine running locally on edge hardware¹. The model is trained on structural e-waste components, including monitor backings, printer fuser panels, personal computer bezels, CRT housings, and laptop enclosures¹. The detection model outputs a bounding box, component label, and confidence score¹. This metadata contextualizes the component within an operational risk matrix: a component classified as a "CRT housing" automatically sets the baseline BFR risk index above 85%, whereas an enclosure identified as a "keyboard casing" defaults to a low-risk ABS profile¹.

Stage 2: Engraved Mold Stamp Optical Character Recognition (OpenCV + EasyOCR)

Under international standards ISO 11469 ("Generic identification and marking of plastics products") and ISO 1043 (Parts 1 to 4: "Symbols and abbreviated terms"), plastic parts over 25 grams must bear an embossed resin identification code⁹. These marks are molded between reversed angle brackets: >ABS<, >PS-HI< (HIPS), >PC<, >PC+ABS<, or >ABS-FR(40)<¹. ISO 1043-4 categorizes flame retardant chemistry using numeric identifiers, where FR(40) indicates halogenated organic compounds with antimony trioxide, and FR(14) or FR(52) designate aliphatic or red phosphorus additives⁹. Because these markings are molded directly into the tooling steel as low-profile geometric reliefs on dark plastic surfaces without color contrast, standard OCR engines fail to read them¹. PolyLoop addresses this issue by routing the cropped region of interest (ROI) through a dedicated OpenCV image processing pipeline prior to character recognition¹.

Processing Sequence	OpenCV Implementation Function	Scientific and Algorithmic Rationale
1. Grayscale Conversion & CLAHE	<pre>cv2.cvtColor(roi, cv2.COLOR_BGR2GRAY) clahe = cv2.createCLAHE(clipLimit=2.5, tileGridSize=(8,8)) enhanced = clahe.apply(gray)</pre>	Standardizes input channels and applies Contrast Limited Adaptive Histogram Equalization, accentuating subtle shadows cast by embossed lettering under directional light ¹ .
2. Bilateral Denoising Filter	<pre>cv2.bilateralFilter(enhanced, d=9, sigmaColor=75, sigmaSpace=75)</pre>	Smooths high-frequency plastic grain noise, scuffs, and mold-texture artifacts while preserving sharp

		gradient transitions across character edges ¹ .
3. Morphological Gradient Extraction	<pre>kernel = cv2.getStructuringElement(cv2.MORPH_RECT, (3, 3)) grad = cv2.morphologyEx(blur, cv2.MORPH_GRADIENT, kernel)</pre>	Subtracts the morphological erosion of the stamp from its dilation, isolating the structural contours and perimeters of the ISO angle brackets ¹ .
4. Otsu Adaptive Binarization	<pre>_, thresh = cv2.threshold(grad, 0, 255, cv2.THRESH_BINARY + cv2.THRESH_OTSU)</pre>	Computes the optimal global binarization threshold based on the bimodal pixel distribution, yielding a high-contrast monochrome image for character parsing ¹ .
5. EasyOCR Inference & Regular Expressions	<pre>results = reader.readtext(thresh) regex = r'>\s*([A-Z0-9\+\-\(\)]+)\s*<'</pre>	Parses text bounding boxes and extracts standardized polymer tokens, automatically flagging embedded safety tags like FR(40) ¹ .

Stage 3: Guided Physical and Chemical Diagnostic Fallback Matrix

When e-waste casings have undergone heavy surface wear, painting, or granulation that obliterates ISO stamps, the system runs an interactive diagnostic wizard¹. The user interface walks the operator through a stepwise physical-chemical screening tree based on density stratification, Hansen solubility reactions, and micro-scale combustion behavior¹.

The density workflow begins by placing clean 5 mm plastic chips into water baths of known specific gravity⁷. Neat polyolefins (PP, LDPE, HDPE) have densities below 0.96 g/cm³ and float in plain tap water ($\rho = 1.00 \text{ g/cm}^3$)⁷. Unfilled HIPS ($\rho = 1.03 - 1.05 \text{ g/cm}^3$) sinks slowly in pure water, but floats cleanly in a 10% sodium chloride (NaCl) saline solution ($\rho = 1.07 \text{ g/cm}^3$)⁷.

Conversely, virgin ABS ($\rho = 1.05 - 1.08 \text{ g/cm}^3$) sinks in a 10% NaCl bath⁷. Pure PC ($\rho = 1.20 - 1.22 \text{ g/cm}^3$) and heavily brominated styrenics (

$\rho = 1.18 - 1.35 \text{ g/cm}^3$) sink rapidly across both baseline liquids due to the high atomic weight of bromine (79.9 g/mol) and antimony (121.8 g/mol)⁶.

For chemical confirmation, a single drop of solvent is applied to an unpainted surface for 30 seconds³. Because of its polar nitrile groups, ABS softens and becomes tacky when exposed to industrial acetone, whereas HIPS exhibits slower surface softening³.

When exposed to *d*-limonene (a bio-derived terpene solvent), HIPS dissolves into a gel, whereas ABS remains unaffected due to the chemical resistance of its SAN copolymer backbone³. Polycarbonate exhibits strong resistance to acetone over short exposures, developing only a faint surface haze after several minutes³.

If the polymer identity remains unconfirmed, a heated copper wire is pressed to the plastic and introduced to a butane flame. The appearance of a bright green flame confirms copper halide formation (chlorine or bromine), indicating hazardous halogenation⁶.

Neat ABS burns with a luminous yellow flame, a blue base, and a pungent odor of burnt rubber¹. HIPS burns rapidly with an orange flame, emitting copious soot webs and a sweet, marzipan-like styrene odor¹. Polycarbonate burns with difficulty, self-extinguishing immediately upon removal from the ignition source while producing a faint phenolic scent¹.

Polyme r Class	Specifi c Gravity (g/cm3)	Tap Water ($\rho=1.00$)	10% NaCl Solutio n ($\rho=1.07$)	Dense Brine ($\rho=1.15$)	Aceton e Spot Reactio n	d-Limo nene Reactio n	Flame and Combu stion Behavi or
HIPS	1.03 – 1.05 ⁷	Sinks slowly ⁷	Floats [cite: 7]	Floats [cite: 10]	Rapidly swells; becom es soft and tacky ³ .	Dissolv es comple tely into a clear, viscous gel ¹³ .	Orange flame, dense soot webs, sweet floral/m arzipan odor ¹ .
ABS	1.05 – 1.08 ⁷	Sinks ⁷	Sinks ⁷	Floats [cite: 10]	Dissolv es rapidly; forms a tacky	Chemic ally inert; zero swelling	Yellow flame, dark smoke, pungen

					paste ³ .	or dissolut ion ¹³ .	t burnt-r ubber odor ¹ .
PC-ABS	1.10 – 1.15	Sinks rapidly	Sinks rapidly	Neutral buoyan cy bounda ry	Slow surface softenin g; cloudy localize d haze ³ .	Chemic ally inert; zero dissolut ion ³ .	Yellow- orange flame, modera te soot, faint aromati c odor ¹ .
PC	1.20 – 1.22 ⁷	Sinks rapidly ⁷	Sinks rapidly ⁷	Sinks rapidly ¹⁰	Highly resistan t; faint micro-c razing after 2 min ³ .	Insolubl e; totally unaffec ted ³ .	Self-ext inguish es upon flame removal ; faint phenoli c odor ¹ .
BFR Plastic (Any + Br)	1.18 – 1.35 [cite: 6]	Sinks rapidly ⁶	Sinks rapidly ⁶	Sinks rapidly ⁶	Variable (govern ed by base resin matrix) ³ .	Variable (govern ed by base resin matrix) ³ .	Self-ex tinguis hes instantl y ; emits dense, choking acid smoke; positive Beilstei n test ⁶ .

Closed-Loop Mechanical Upcycling and 3D Filament Extrusion Parameters

Transforming post-consumer casings into 1.75 mm additive manufacturing filament requires strict rheological and moisture control¹. Recycled polymers undergo multiple thermal processing cycles, and contaminated feedstocks or incorrect processing temperatures

degrade polymer chains, resulting in brittle, unprintable filament¹.

Granulation and Hygroscopic Moisture Extraction

Recovered enclosures approved by the BFR Risk Engine are manually stripped of copper wiring, brass thread inserts, and foam padding, then processed through a rotary granulator to produce 3 mm to 5 mm regrind flakes¹. Fine dust and particulate matter are removed through a 1 mm mesh sieve¹.

Engineering thermoplastics—especially Polycarbonate and ABS—are hygroscopic, readily absorbing ambient atmospheric moisture into their polymer matrices¹. When moist polymer flakes enter the heated barrel of an extruder, the trapped water flashes into high-pressure steam¹. In Polycarbonate and PC-ABS blends, this triggers hydrolytic chain scission: water molecules chemically cleave the ester linkages of the carbonate backbone at elevated

temperatures, permanently reducing average molecular weight ($\overline{M}_w$) and intrinsic viscosity¹. The resulting extruded filament exhibits internal voiding, surface foaming, and extreme brittleness¹.

Polymer Regrind Type	Minimum Pre-Drying Temp	Required Residence Time	Maximum Permissible Moisture	Extrusion Failure Mode if Undried
Recycled ABS	80°C (176°F)	3 to 4 hours	< 0.05% by weight (500 ppm)	Steam bubbling in melt die, rough matte finish, severe layer shear weakness ¹ .
Recycled HIPS	70°C (158°F)	2 to 3 hours	< 0.05% by weight (500 ppm)	Micro-voiding, nozzle spitting, and inconsistent filament diameter ¹ .
Recycled PC	120°C (248°F)	4 to 6 hours	< 0.02% by weight (200 ppm)	Catastrophic hydrolytic chain scission, permanent embrittlement, structural

				failure ¹ .
Recycled PC-ABS	100°C (212°F)	3 to 4 hours	< 0.03% by weight (300 ppm)	Internal foaming, poor interlayer bonding, delamination under minimal flexure ¹ .

Extrusion Rheology and 3D Printing Operational Parameters

The dried flakes are processed using a single-screw filament extruder equipped with a compression screw featuring a 3:1 length-to-diameter (L/D) compression ratio¹. The extruder barrel is divided into three independently controlled heating zones, terminating in a 1.68 mm melt die orifice that expands under polymer die swell to yield a nominal 1.75 mm (± 0.05 mm) filament diameter¹. Filament tension is regulated by a dual-axis optical laser diameter gauge linked to a variable-speed tractor puller spooled directly onto 1 kg reels¹. The table below outlines validated operating boundaries for single-screw extrusion and end-use Fused Deposition Modeling (FDM) additive manufacturing:

Processing Parameter	Recycled ABS	Recycled HIPS	Recycled PC	Recycled PC-ABS Blend
Extruder Feed Zone 1	185°C – 195°C	170°C – 180°C	230°C – 245°C	210°C – 220°C
Extruder Transition Zone 2	215°C – 225°C	195°C – 210°C	265°C – 275°C	235°C – 245°C
Extruder Die Zone 3	225°C – 235°C	210°C – 225°C	280°C – 295°C	250°C – 265°C
Cooling Water Bath Temp	50°C – 60°C (warm bath)	40°C – 50°C	70°C – 80°C	60°C – 70°C
FDM Printing Nozzle Temp	230°C – 245°C	220°C – 240°C	280°C – 305°C (All-Metal)	255°C – 275°C

FDM Heated Bed Temp	95°C – 110°C	85°C – 100°C	115°C – 130°C	100°C – 115°C
Build Plate Interface	Smooth PEI + Dimafix adhesive	PEI Sheet or Kapton Tape	PEI + Layered PVA Glue Stick	Textured PEI Sheet
Chamber Enclosure Temp	Passive enclosure (40°C – 50°C)	Passive enclosure (35°C – 45°C)	Actively heated (65°C – 80°C)	Enclosed chamber (45°C – 60°C)
Part Cooling Fan Speed	0% – 15% (preserves adhesion)	20% – 35%	0% (zero draft to prevent stress)	10% – 20%
Volumetric Thermal Shrinkage	0.4% – 0.7% (high warp risk)	0.3% – 0.6% (moderate warp risk)	0.6% – 0.9% (severe warp risk)	0.5% – 0.7% (moderate warp risk)

Occupational Health, Safety, and Volatile Organic Compound (VOC) Control

Extruding and printing recycled styrenics and engineering polymers generates hazardous airborne emissions¹. Thermally processed ABS and HIPS release volatile organic compounds (VOCs)—predominantly styrene monomer, ethylbenzene, and acrylonitrile—alongside high concentrations of Ultrafine Particles (UFPs, aerodynamically $< 100 \text{ nm}$)¹. Styrene is an occupational neurotoxin and an IARC Group 2A probable human carcinogen¹. Extrusion equipment and 3D printers must operate within sealed enclosures linked to local exhaust ventilation systems equipped with activated carbon adsorptive beds for VOC capture and HEPA filters rated to trap particulates down to 0.3 microns¹.

Carbon Avoidance Life Cycle Assessment (LCA)

Methodology

To quantify environmental benefits, PolyLoop incorporates an automated, ISO 14040/14044-compliant Life Cycle Assessment (LCA) computational engine¹. The model benchmarks mechanical upcycling directly against primary production of virgin petrochemical resins¹.

Mathematical Formulation

The net Global Warming Potential (GWP) offset achieved per functional unit of processed e-waste plastic is calculated via:

$$\Delta E_{\text{avoided}} = M_{\text{recycled}} \cdot (EF_{\text{virgin}} - EF_{\text{recycling}}) + M_{\text{diverted}} \cdot EF_{\text{landfill_credit}} - M_{\text{recycled}} \cdot EF_{\text{transport}}$$

Where:

- M_{recycled} represents the total mass of clean, verified polymer successfully extruded into commercial-grade 3D filament (kg)¹.
- EF_{virgin} represents the cradle-to-gate Global Warming Potential of manufacturing equivalent virgin resin (kg CO₂e per kg resin), sourced from PlasticsEurope Eco-Profiles and Ecoinvent v3.8¹.
- $EF_{\text{recycling}}$ represents the cumulative energy expenditure of mechanical sorting, grinding, washing, convective air-drying, and single-screw thermal extrusion (empirically established at 0.75 kg CO₂e/kg based on regional grid electricity mix)¹.
- $EF_{\text{landfill_credit}}$ quantifies the baseline avoidance of fugitive methane, volatile organic releases, and heavy metal leachate treatment resulting from landfill diversion (fixed at 0.05 kg CO₂e/kg)¹.
- $EF_{\text{transport}}$ accounts for local reverse logistics, collection, and transport fuel emissions (calculated at 0.08 kg CO₂e/kg over a 50 km logistics radius)¹.

Polymer Classification	Virgin Cradle-to-Gate GWP (EF _{virgin})	PolyLoop Process Burden (EF _{recycling} + EF _{transport})	Net Carbon Avoided (ΔE _{avoided})	Equivalent Fossil Crude Oil Offset
ABS (Acrylonitrile Butadiene Styrene)	3.80 kg CO ₂ e / kg ¹⁸	0.83 kg CO ₂ e / kg ¹	+ 3.02 kg CO₂e / kg [cite: 1]	~ 2.1 Liters crude oil equivalent ¹
HIPS (High-Impact Polystyrene)	3.35 kg CO ₂ e / kg ¹⁹	0.83 kg CO ₂ e / kg ¹	+ 2.57 kg CO₂e / kg [cite: 1]	~ 1.8 Liters crude oil equivalent ¹

PC (Polycarbonat e)	6.80 kg CO ₂ e / kg ²⁰	0.83 kg CO ₂ e / kg ¹	+ 6.02 kg CO₂e / kg [cite: 1]	~ 3.9 Liters crude oil equivalent ¹
PC-ABS Blend	5.10 kg CO ₂ e / kg ¹⁹	0.83 kg CO ₂ e / kg ¹	+ 4.32 kg CO₂e / kg [cite: 1]	~ 2.9 Liters crude oil equivalent ¹

Empirical Batch Simulation (100 kg E-Waste Plastic Inflow)

To evaluate system performance at pilot scale, a 100 kg batch of mixed commercial IT scrap was analyzed¹:

- An incoming mix of 100 kg containing computer bezels, monitor frames, and printer covers enters sorting¹.
- The BFR Risk Engine identifies and isolates 15 kg of older CRT monitor casings containing over 2,000 ppm PBDE, diverting them from thermal extrusion to compliant hazardous containment¹.
- The remaining 85 kg of clean material is sorted into 50 kg ABS, 25 kg PC-ABS, and 10 kg HIPS¹.
- Calculating net carbon avoidance:

$$\Delta E = (50 \times 3.02) + (25 \times 4.32) + (10 \times 2.57) = 151.0 + 108.0 + 25.7 = 284.7 \text{ kg CO}_2\text{e}$$

This single 100 kg recovery run eliminates 284.7 kg of net greenhouse emissions, offsetting the environmental impact of combusting 140 kg of coal or operating an internal combustion vehicle for roughly 1,160 kilometers¹.

100% Free Open-Source Architecture and Tech Stack

PolyLoop is designed to operate without paid cloud infrastructure, proprietary vision APIs, or subscription databases¹. The architecture executes locally on standard consumer PC hardware using open-source tools¹.

The system architecture organizes data flow across three primary layers:

1. **Presentation Layer:** The user interface runs as a single-page application built with React 18, Vite, and Tailwind CSS¹. It provides a video canvas targeting overlay, interactive diagnostic forms, telemetry dashboards using Recharts, and direct triggers for Material Passport PDF generation¹.
2. **Application and Logic Layer:** A local FastAPI server processes incoming requests¹. Visual classification is handled by an edge-optimized Ultralytics YOLOv8n model, while engraved mold markings are parsed using OpenCV preprocessing filters and EasyOCR¹. Deterministic business logic is divided into modular services: a Polymer Knowledge Base defining physical-chemical constants, a BFR Contamination Risk Engine evaluating device

age and safety codes, a Thermal Parameter Calculator generating machine settings, and an ISO 14040 Carbon Avoidance Engine computing environmental savings¹.

3. **Data and Output Layer:** Session states, component scan histories, and audit records are stored locally in an embedded SQLite database managed via SQLAlchemy¹. The system uses ReportLab to compile multi-page, verifiable Digital Material Passports directly on the backend, generating downloadable PDF certificates with complete chain-of-custody data¹.

Architectural Domain	Selected Technology	Open-Source License	Functional Role and Zero-Cost Justification
Vision & Object Detection	Ultralytics YOLOv8n / YOLOv11n ¹	AGPL-3.0 / Apache 2.0 ¹	Local CPU/GPU inference for casing classification without cloud vision API costs ¹ .
Optical Character Recognition	EasyOCR + PyTorch ¹	Apache 2.0 ¹	Local character recognition engine for parsing ISO resin stamps on ROIs ¹ .
Image Preprocessing	OpenCV (opencv-python-headless) ¹	Apache 2.0 ¹	Local image operations: CLAHE contrast enhancement, bilateral filtering, Otsu binarization ¹ .
API Web Service	FastAPI + Pydantic v2 ¹	MIT ¹	Asynchronous Python framework with automatic Swagger documentation and schema validation ¹ .
Data Persistence	SQLite + SQLAlchemy ORM ¹	Public Domain / MIT ¹	Serverless, zero-configuration local relational

			database stored in a versioned flat file ¹ .
Frontend Framework	React 18 (Vite build engine) ¹	MIT ¹	High-performance SPA with fast hot module replacement and minimal bundle size ¹ .
Styling & Icons	Tailwind CSS + Lucide React ¹	MIT ¹	Utility-first responsive design; dark-mode industrial theme; comprehensive icon set ¹ .
Data Visualization	Recharts ¹	MIT ¹	Declarative SVG charting library for thermal curves, batch breakdowns, and LCA metrics ¹ .
Document Generation	ReportLab Document Engine ¹	BSD License ¹	Programmatic PDF compilation; produces downloadable technical Material Passports ¹ .
DevOps & Integration CI	GitHub Actions ¹	Free Public Tier ¹	Automated continuous integration: executes Ruff linting and Pytest test suites on PRs ¹ .

Four-Member Engineering Division, Repository Governance, and Sprint Roadmap

To ensure parallel development across a four-engineer team, PolyLoop establishes clear operational boundaries, interface contracts, and GitFlow governance rules¹.

Team Role Allocation and API Interface Contracts

Engineering responsibilities are divided into four specialized domains¹:

- **Member 1 (Vision and OCR Pipeline Lead):** Focuses on computer vision, YOLO model optimization, OpenCV image filtering, and EasyOCR stamp extraction¹. Member 1 ingests raw camera frames and delivers a standardized JSON payload: { casing_type: str, stamp_detected: bool, raw_text: str, polymer: str, confidence: float }¹.
- **Member 2 (Backend Architecture and Domain Lead):** Responsible for API routing, the Polymer Knowledge Base dictionary, BFR risk-scoring algorithms, and SQLite database schemas¹. Member 2 ingests the vision payload and diagnostic inputs to provide unified REST endpoints (/api/scan, /api/diagnose, /api/batches)¹.
- **Member 3 (Frontend UI and Diagnostic Wizard Lead):** Implements the React single-page application, WebRTC camera canvas overlay, diagnostic decision tree, and Recharts telemetry dashboards¹. Member 3 delivers an industrial user interface connected to the backend API¹.
- **Member 4 (Circularity, Thermal Modeling, PDF and DevOps Lead):** Manages extrusion parameter calculations, the ISO 14040 carbon offset engine, ReportLab PDF passport generation, and GitHub Actions CI pipelines¹. Member 4 ingests verified batch records to output PDF technical datasheets (/api/passport/{id}/pdf) and maintains repository test coverage¹.

Team Role & Specialization	Concrete Deliverables	Input Interface Contract	Output Interface Contract
Member 1: Vision & OCR Pipeline Lead ¹	• Quantized YOLOv8n casing classifier¹. • OpenCV adaptive thresholding pipeline¹. • EasyOCR wrapper with ISO regex parser¹. • Standalone Python inference module¹.	Raw webcam video stream, captured snapshots, and image bytes ¹ .	Standardized JSON payload: { casing: str, stamp: bool, text: str, polymer: str, conf: float } ¹ .
Member 2:	• FastAPI server	Vision JSON	REST endpoints:

Backend & Polymer PKB Lead ¹	routing, CORS, and schemas¹. • Polymer Knowledge Base property dictionary¹. • BFR historical risk-scoring algorithms¹. • SQLite scan persistence layer¹.	payloads from Member 1 and test inputs from Member 3 ¹ .	POST /api/scan, POST /api/bfr-risk, GET /api/batches/{id} ¹ .
Member 3: Frontend UI & Diagnostic HUD Lead ¹	• Responsive React 18 SPA (Vite + Tailwind)¹. • WebRTC camera component with canvas overlay¹. • Step-by-step diagnostic questionnaire UI¹. • Real-time Recharts telemetry graphs¹.	User click inputs, video feed, and API responses from Member 2 ¹ .	Responsive industrial web interface with integrated PDF download triggers ¹ .
Member 4: LCA, Extrusion Math & DevOps Lead ¹	• Filament extrusion thermal calculation logic¹. • ISO 14040 carbon avoidance math engine¹. • ReportLab automated PDF	Verified polymer classifications and batch weights from Member 2 ¹ .	Downloadable PDF Material Passports and passing automated CI test suites ¹ .

	passport script ¹ . • GitHub Actions CI pipeline (Ruff + Pytest) ¹ .		
--	---	--	--

Ten-Day Sprint Execution Schedule

Development is structured into three consecutive sprint milestones¹:

- **Sprint 1 (Days 1 – 3: Foundations and Scaffolding):** Member 1 curates a reference dataset of 25 casing stamp images and tests OpenCV CLAHE filters¹. Member 2 scaffolds the FastAPI codebase and implements the static Polymer Knowledge Base dictionary¹. Member 3 initializes the Vite and Tailwind workspace, building the shell layout, header, and tab navigation¹. Member 4 establishes the carbon offset equations, sets up repository branch protections, and configures GitHub Actions CI workflows¹.
- **Sprint 2 (Days 4 – 7: Core Feature Implementation):** Member 1 refines EasyOCR extraction and implements ISO 11469 regex matching¹. Member 2 builds the BFR risk-scoring algorithm and SQLite persistence tables¹. Member 3 creates the HTML5 camera canvas HUD and the interactive sink-float questionnaire flow¹. Member 4 develops the thermal parameter calculator and designs the ReportLab PDF passport template¹.
- **Sprint 3 (Days 8 – 10: System Integration and Demo Freeze):** Member 1 packages the vision scripts into an API-callable Python module¹. Member 2 connects OCR and BFR logic into the master /api/scan endpoint¹. Member 3 connects the frontend video feed to backend routes, binding live results to UI cards and passport download buttons¹. Member 4 executes end-to-end integration tests, verifies PDF output formatting, and tags the final production release¹.

Milestone Timeline	Member 1 (Vision/OCR)	Member 2 (Backend/PKB)	Member 3 (Frontend/HU D)	Member 4 (LCA/DevOps)
Sprint 1: Days 1 – 3 <i>Foundational Architecture</i> [cite: 1]	Curate 25 casing stamp images; validate OpenCV CLAHE filter scripts ¹ .	Scaffold FastAPI application; construct static PKB dictionary ¹ .	Initialize Vite+Tailwind workspace; build shell layout and tabs ¹ .	Formulate LCA carbon math; configure GitHub Actions workflows ¹ .
Sprint 2: Days	Finalize	Build BFR risk	Develop	Build thermal

4 – 7 <i>Core Logic Implementation</i> [cite: 1]	EasyOCR extraction; implement ISO 11469 regex parser ¹ .	scoring logic; construct SQLite schema and CRUD routes ¹ .	WebRTC camera HUD; implement interactive test wizard ¹ .	parameter calculator; design ReportLab PDF template ¹ .
Sprint 3: Days 8 – 10 <i>Integration & Demo Freeze</i> [cite: 1]	Package vision pipeline into clean, importable Python modules ¹ .	Wire OCR, PKB, and BFR logic into master /api/scan endpoint ¹ .	Connect video streams to backend; bind passport download triggers ¹ .	Run end-to-end integration tests; resolve CI warnings; tag v1.0 release ¹ .

Repository Governance and Pull Request Quality Gates

Development follows a strict GitFlow workflow:

- The main branch is production-frozen, reserved exclusively for tagged, demo-ready releases¹.
- The develop branch serves as the staging branch for feature integration; direct commits are prohibited¹.
- All feature implementations are isolated in dedicated branches named by member role (e.g., feat/m1-ocr-stamp-extractor, feat/m2-bfr-risk-scoring, feat/m3-sink-float-wizard-ui, feat/m4-pdf-passport-generator)¹.
- Pull Request Approval Criteria: Merging into develop requires at least one peer code review, clean test execution via pytest without regressions, zero warnings under the ruff linter, successful compilation under npm run build, and complete absence of hardcoded commercial API keys¹.

Plain-Language Synthesis: Core Problem, Operational Functionality, and Societal Impact

The technical specifications of the PolyLoop system can be summarized through four core questions:

1. What Is the Underlying Problem with E-Waste Plastics?

When electronics like computers, monitors, and printers reach the end of their usable life, recyclers systematically recover valuable metals such as gold, silver, and copper from circuit boards¹. However, the plastic casings—which make up nearly a third of the machine's total mass—are almost universally thrown away¹.

This systemic waste occurs for two main reasons:

1. **The Incompatibility Problem:** Different plastics look identical¹. A black monitor bezel made of Acrylonitrile Butadiene Styrene (ABS) looks and feels just like one made of

High-Impact Polystyrene (HIPS)¹. However, if an operator melts them together, they do not mix; they separate on a microscopic level, producing a fragile material that crumbles under minimal stress¹.

2. **The Toxic Contamination Problem:** Millions of older electronics contain Brominated Flame Retardants (BFRs)—hazardous chemical additives designed to prevent fires¹. When remelted, these plastics release toxic dioxins, making them unsafe and illegal under international environmental laws (such as RoHS and WEEE) for use in consumer products, toys, or 3D printing filaments¹.

Because small-scale recyclers, maker spaces, and municipal centers lack access to expensive industrial analytical laboratories to sort these plastics safely, over 80% of e-waste plastic is sent to landfills or incinerators¹.

2. What Does the PolyLoop System Actually Do?

PolyLoop is an open-source, digital materials platform that runs on a standard laptop equipped with a basic webcam¹. It serves as an automated sorting and verification assistant through a four-step workflow:

1. **Automated Visual Recognition:** An operator holds a scrap plastic casing up to the webcam¹. The computer vision system identifies what kind of appliance it is (e.g., a printer door, a computer chassis, or a CRT housing)¹.
2. **Mold Stamp Reading:** The camera focuses on the plastic's interior surface, using image-filtering algorithms to locate and read tiny manufacturer mold markings (such as >ABS< or >PC+ABS<) that are embossed directly into the material¹.
3. **Physical-Chemical Diagnostic Fallback:** If the plastic is scratched, painted, or broken and has no legible stamp, the system guides the operator through a quick test using common household liquids¹. By observing whether a small scrap floats or sinks in plain tap water or salt water, and checking how it reacts to a single drop of citrus oil (*d*-limonene) or acetone, the system identifies the exact polymer class in under two minutes¹.
4. **Safety and Hazard Screening:** The system calculates a contamination risk score based on the component's age and device category¹. Casings flagged for high flame-retardant concentrations are safely rejected from recycling streams¹.

3. What Downstream Solution Does PolyLoop Provide?

When a plastic sample is confirmed pure and non-toxic, PolyLoop generates precise processing instructions¹. It outputs the exact pre-drying times, shredding requirements, and multi-zone heating temperatures needed to melt down the scrap and extrude it into high-grade 1.75 mm 3D printer filament¹.

Simultaneously, the system calculates the environmental savings achieved by diverting that specific batch from a landfill and displacing virgin fossil fuels, automatically compiling these metrics into a downloadable, certified Digital Material Passport PDF¹.

Operational Dimension	Conventional Disposal Paradigm	PolyLoop Closed-Loop Upcycling Paradigm
Material Identification	Casings appear identical; sorting requires industrial NIR/XRF spectroscopy costing over \$30,000 ¹ .	Uses local computer vision, automated ISO mold stamp OCR, and guided benchtop density/solvent diagnostics ¹ .
Chemical Safety Assurance	BFR-doped plastics are commingled into unsafe recycled products or landfilled indiscriminately ¹ .	Evaluates age and appliance typology to identify and isolate halogenated flame retardants before remelting ¹ .
Mechanical Integrity	Incompatible plastics are cross-contaminated, causing extreme brittleness and structural delamination ¹ .	Segregates pure resin batches (ABS, HIPS, PC), preserving mechanical strength and layer bonding ¹ .
Software Infrastructure	Dependent on proprietary cloud models, monthly API keys, or specialized external services ¹ .	Built on a 100% free, locally executed open-source stack (YOLO, OpenCV, FastAPI, SQLite, React, ReportLab) ¹ .
Economic Value Realization	E-waste plastic remains a disposal cost, ending up buried in landfills or burned in municipal incinerators ¹ .	Upcycles discarded scrap into 3D printer filament worth \$20 to \$40 per kilogram ¹ .
Circularity Traceability	Zero documentation on recycled polymer purity, chain of custody, or carbon offsets ¹ .	Generates a tamper-evident, downloadable Material Passport PDF logging batch purity and net CO ₂ e savings ¹ .

PolyLoop bridges the gap between electronic waste disposal and decentralized additive manufacturing¹. By replacing capital-intensive analytical hardware with open-source software,

image processing, and basic chemistry, the platform enables municipal facilities, educational institutions, and independent makers to safely transform hazardous electronic trash into a verified, circular manufacturing feedstock¹.

Works cited

1. polyloop - ewem green hackathon.pdf
2. Ecotoxicity of Plastics from Informal Waste Electric and Electronic, <https://pmc.ncbi.nlm.nih.gov/articles/PMC7712128/>
3. Chemical Recycling of Mixed Plastics in Electronic Waste Using, <https://psecommunity.org/wp-content/plugins/wpor/includes/file/2302/LAPSE-2023.3157-1v1.pdf>
4. A Review of Methods and Data on the Recycling of Plastics ... - MDPI, <https://www.mdpi.com/2313-4321/10/4/148>
5. Review of Reagent-Free Electronic Waste Recycling - MDPI, <https://www.mdpi.com/2313-4321/11/2/27>
6. Detection of Brominated Plastics from E-Waste by Short-Wave, <https://www.mdpi.com/2313-4321/6/3/54>
7. (PDF) Separation of virgin plastic polymers and post-consumer, https://www.researchgate.net/publication/353715618_Separation_of_virgin_plastic_polymers_and_post-consumer_mixed_plastic_waste_by_sinking-flotation_technique
8. Recycling of plastics with flame retardants of electronic waste, a, https://www.reward-weee.eu/downloads/EGG2012_Berlin_EFRA_REWARD_Recycling_Brominated_Plastics.pdf
9. ISO 11469 - iTeh Standards, <https://cdn.standards.itih.ai/samples/27946/5c3ba9b34fca4f4aaa7043e014810004/ISO-11469-2000.pdf>
10. Recovery of Plastics from WEEE through Green Sink–Float Treatment, <https://www.mdpi.com/1996-1944/17/12/3041>
11. INTERNATIONAL STANDARD ISO 11469, <https://cdn.standards.itih.ai/samples/63434/f460de8a1261420a92f1195a8e329213/ISO-11469-2016.pdf>
12. Plastic Part Marking – Overview Codes and Standards (incl, <https://www.findoutaboutplastics.com/2020/12/plastic-part-marking-overview-codes-and.html>
13. ASA vs HIPS: Outdoor Durability, Impact Resistance and Support Use, <https://filamentcompare.com/asa-vs-hips/>
14. The Product Carbon Footprint Guideline for the Chemical Industry, https://www.tfs-initiative.com/app/uploads/2024/03/TfS_PCF_guidelines_2024_EN_pages-low.pdf
15. Eco-profiles program and methodology PlasticsEurope Version 3.0, https://plasticseurope.org/wp-content/uploads/2021/12/PlasticsEurope-Ecoprofiles-program-and-methodology_V3.0.pdf
16. The impact of plastics on life cycle energy consumption and,

<https://plasticseurope.org/wp-content/uploads/2021/10/201009-Denkstatt-Report.pdf>

17. Fundamental, technical and environmental overviews of plastic,
<https://pubs.rsc.org/gc/article/26/23/11444/851968/Fundamental-technical-and-environmental-overviews>
18. Plastberegner.dk is developed by 2. - 2-0 LCA,
https://lca-net.com/files/Plastindustrien_LCAreport_20121008.pdf
19. Mandatory percentage of recycled or bio-based plastic - CE Delft,
https://cedelft.eu/wp-content/uploads/sites/2/2022/03/CE_Delft_200289_Mandatory_percentage_of_recycled_or_bio-based_plastic_Def.pdf
20. From Life Cycle Assessment to - RWTH Publications,
<https://publications.rwth-aachen.de/record/988442/files/988442.pdf>
21. PAO Portable Lamp - Design Life-Cycle,
<http://www.designlife-cycle.com/pao-portable-lamp>
22. [unknown_url](#)